Machine Learning-Based House Price Prediction

Feature Engineering, Model Optimization & Performance Comparison | AI & ML Track

Introduction:

This task builds on the Linear Regression model developed in Task 1 by comparing multiple machine learning models and evaluating their performance on the California Housing dataset. The dataset contains 20,640 samples and 8 input features used to predict median house values. Before training, feature scaling was applied using StandardScaler to bring all features to a similar range and improve model consistency. The dataset was then divided into training and testing sets using an 80:20 split with a random state of 42. Three regression models were trained and evaluated:

- **Linear Regression:** Used as the baseline model for comparison.
- **Ridge Regression (alpha = 1.0)**: Introduces L2 regularisation to improve model stability and reduce the risk of overfitting.
- **Decision Tree Regressor (max_depth = 5):** Capable of capturing non-linear relationships between features and house prices.

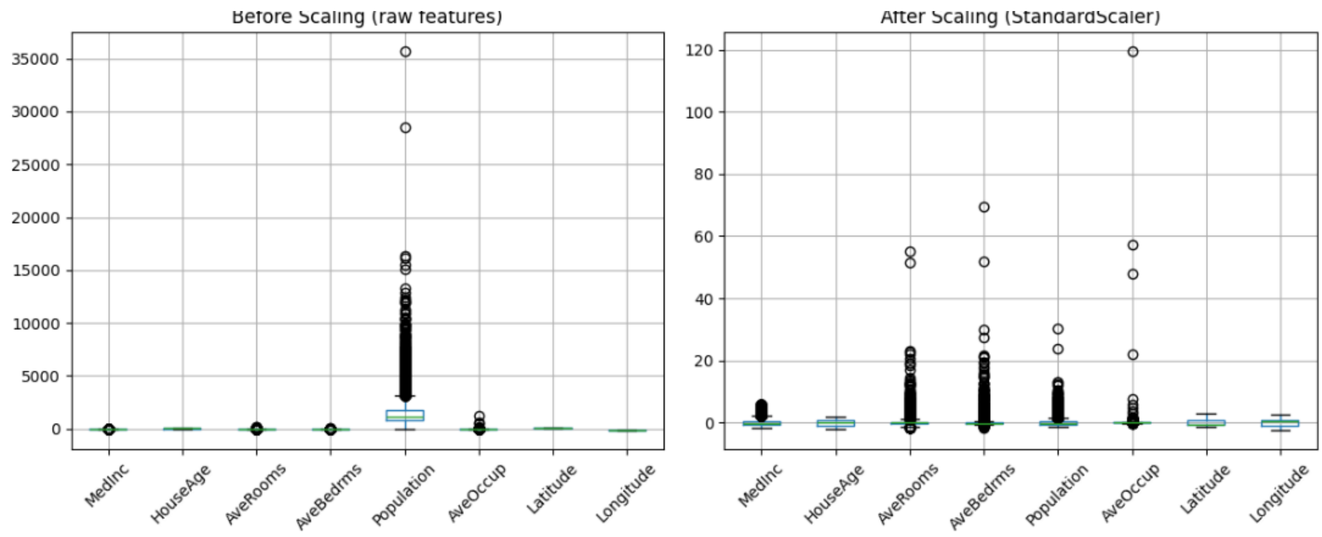
The performance of each model was compared using MAE, RMSE, and R^2 Score to determine the most effective model for house price prediction.

Dataset Overview:

The California Housing dataset was used for this project. It contains 20,640 samples and 8 numerical features related to housing and demographic information. The target variable represents the median house value for each district. The dataset includes features such as median income, house age, average number of rooms, average number of bedrooms, population, occupancy, latitude, and longitude. These features were used to predict house prices and compare the performance of different regression models.

Feature Scaling Effect:

The plots show the distribution of features before and after applying StandardScaler. Before scaling, the features had significantly different ranges. After scaling, all features were standardized to a similar scale, improving consistency and enabling a fair comparison between the machine learning models.



Model Training:

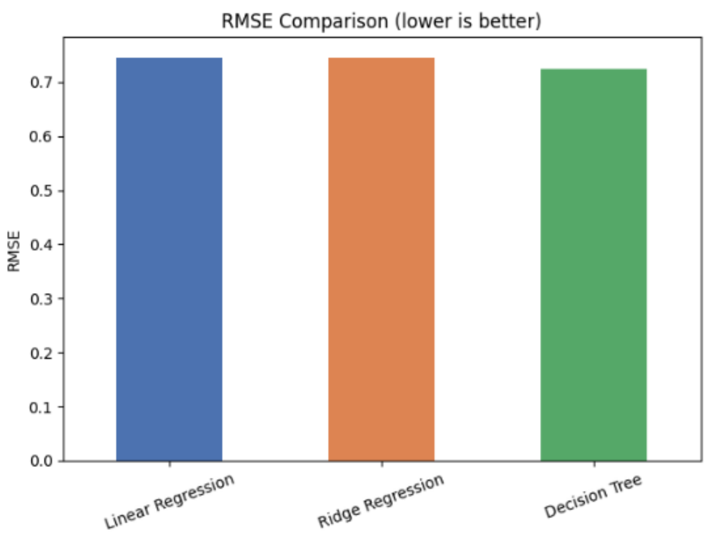
After feature scaling, the dataset was divided into training and testing sets using an 80:20 split. Three regression models were trained on the training data: Linear Regression, Ridge Regression, and Decision Tree Regressor. Linear Regression was used as the baseline model, while Ridge Regression was included to evaluate the effect of regularisation. A Decision Tree Regressor with a maximum depth of 5 was also trained to capture non-linear relationships within the data. The trained models were then evaluated on the test set using MAE, RMSE, and R^2 Score.

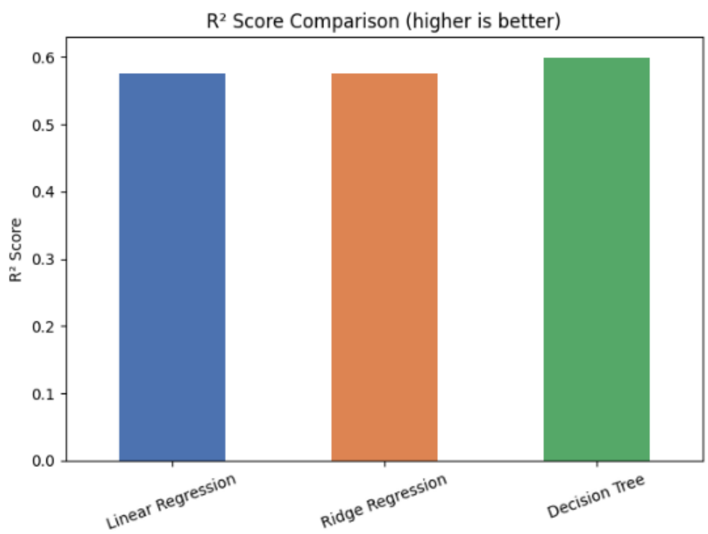
Model Performance Comparison:

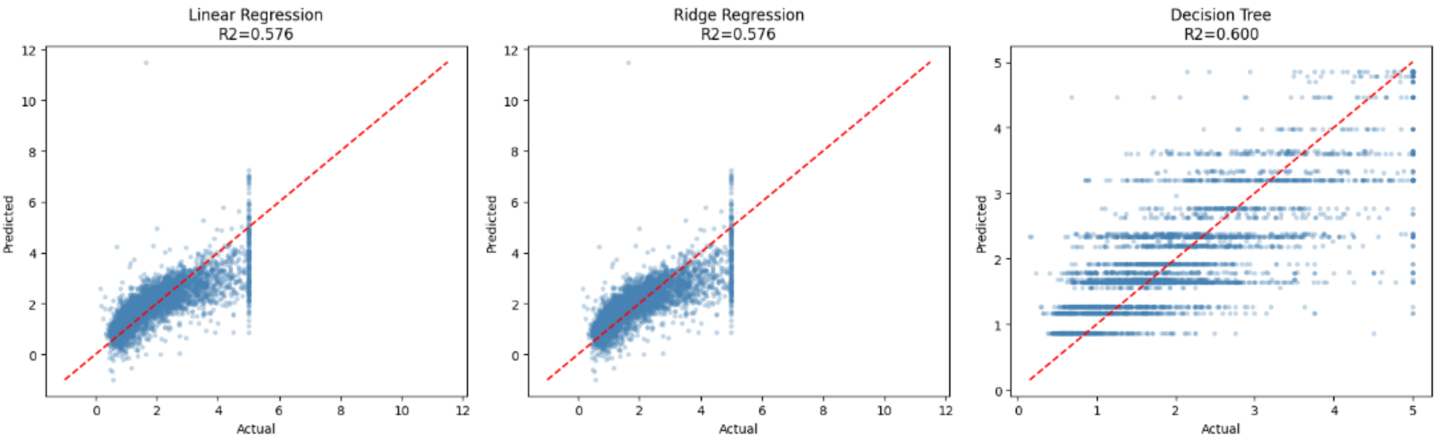
The performance of the three regression models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Lower MAE and RMSE values indicate better prediction accuracy, while a higher R^2 Score indicates a better fit to the data.

Among the evaluated models, the Decision Tree Regressor achieved the best overall performance with the lowest error values and the highest R^2 Score. Linear Regression and Ridge Regression produced very similar results, while the Decision Tree was able to capture patterns in the dataset more effectively. Based on these results, the Decision Tree Regressor was selected as the best-performing model.

Model	MAE	RMSE	R^2 Score
Linear Regression	0.5332	0.7456	0.5758
Ridge Regression	0.5319	0.7456	0.5758
Decision Tree (depth=5)	0.5223	0.7242	0.5997






Linear Regression
R2=0.576
Ridge Regression
R2=0.576
Decision Tree
R2=0.600
Predicted
Actual


Best Model: Decision Tree (R2=0.600)
Perfect Prediction
Predicted House Price
Actual House Price

Results & Interpretation:

- Linear Regression and Ridge Regression produced very similar results, indicating that regularisation had little impact on model performance for this dataset. This suggests that the baseline linear model was not significantly affected by overfitting.
- The Decision Tree Regressor achieved the best performance among the three models, with lower error values and a higher R^2 Score. Its improved performance suggests that the relationship between housing features and house prices is not entirely linear. By capturing more complex patterns in the data, the Decision Tree was able to make more accurate predictions than the linear models.

Final Model Selection & Justification:

The Decision Tree Regressor was selected as the final model because it achieved the best overall performance among the evaluated models. It produced the lowest prediction error and the highest R^2 Score, indicating a better fit to the data. The chosen maximum depth of 5 helps keep the model relatively simple while still capturing important patterns within the dataset.

Conclusion:

In this task, multiple regression models were trained and compared using the California Housing dataset. After applying feature scaling and evaluating model performance using MAE, RMSE, and R^2 Score, the Decision Tree Regressor emerged as the best-performing model.

This project highlighted the importance of comparing different machine learning models rather than relying on a single approach. Future improvements could include hyperparameter tuning, cross-validation, and testing more advanced models to further improve prediction accuracy.